

Here are some nominal wh-clauses in Greek*

Christos Vlachos
University of Patras
cvlachos@upatras.gr

Abstract

This paper discusses novel empirical evidence from Greek showing that a wh-clause (i.e., a CP) may correspond to a nominal constituent with referential interpretation (just like a DP). The evidence comes from the distribution of wh-clauses with the presentative particle *nà* (roughly translating to the English *here is/are*, the Italian *ecco*, or the French *voilà*). *nà* places two restrictions on the type of complement it may take, one relating to syntax and the other to the semantics: The complement must be ‘nominal’ in nature and must bear a referential reading. After showing that a wh-clause combined with *nà* is a complement and not an adjunct, I propose that the nominal status of the wh-clause comes from its D-label. This label, in turn, results from the association of the wh-phrase with C, on the following two assumptions (both of which are established elsewhere, while I further elaborate here on the second): (i) Greek wh-phrases, regardless of the grammatical category they belong to (i.e., N, Adj., Adv.), are (bare) indefinite D-elements (and not intrinsically interrogative quantifiers); and (ii) C bears a generalized Op(erator) feature that requires a value and assigns quantificational properties to the element it combines with (here, the wh-phrase). More generally, this analysis raises important, and far-reaching, implications vis-à-vis the syntax and semantics of wh-clauses, clause-typing, and clausal complement selection, on which I barely touch at present.

Keywords: wh-clause; labeling; presentative; complementation; selection

1. Introduction

A long-standing strand of research on clausal complementation has pursued the hypothesis that a clausal constituent may have a nominal external syntax, whereby ‘external’ I mean the outmost left-peripheral layer of a constituent (e.g., Kanye 1982; Adger & Quer 2001; Kayne 2008). This hypothesis, which can be traced back to Rosenbaum (1967) and Kiparsky & Kiparsky (1971), is empirically based mostly on declarative clauses and to some extent on relative clauses and polar questions. In turn, this empirical focus stems from the observation that complementizers correlate with pronominal elements, to the extent that some authors take complementizers not to form a distinct categorial class but to reflect grammatical functions of (pro)nominal elements (e.g., Manzini 2010; Roussou 2010). Under this perspective, wh-clauses have mostly gone under the radar because wh-clauses are introduced by wh-phrases and the latter do not typically function as complementizers. Yet, even in the case of wh-clauses, we do find instances of syncretism between complementizers and pronouns, which alludes to some sort of nominal syntax. A recent example is Manzini & Roussou (2024), who examine cases of syncretism between the Italian complementizer *that* and the wh-phrase for *what*, both of which are manifested by the Italian pronoun *che*.

In this paper, I wish to probe further into the external syntax of wh-clauses and argue that a wh-clause may correspond to a nominal element.¹ In doing so, I present empirical evidence from (Modern) Greek that have not been discussed so far (as I know) and can be exemplified as in (1).

* This paper has been presented at the workshop “Information Structure in Greek: Interface and comparative studies”, in the context of the 16th International Conference on Greek Linguistics (ICGL 16). For helpful comments and discussions, I am indebted to the participants of the workshop, and more in particular, Dora Alexopoulou, Nikos Angelopoulos, Richard Faure, Konstantia Kapetangianni, George Kotzoglou, Dimitris Michelioudakis and George Tsoulas. Moreover, I want to thank the members of the Patras Syntax and Semantics (PaSS) reading group for useful suggestions at an earlier stage of development of the research reported here. Special thanks goes to Michalis Georgiafentis, Stavros Skopeteas and Angeliki Tsokoglou, for organizing the workshop and putting together this special issue in which the present paper appears. Last, but not least, I cordially thank two anonymous reviewers for their constructive criticism and food for (further) thought. I take full responsibility of all remaining errors.

¹ See also Vlachos & Balasi (2022) for the same conclusion drawn from different empirical evidence.

- (1) *Nà pjos tha erthi me mas.*
 here-is who.NOM will come.3SG with us
 “Here is the person that will come with us.”

Let us put (1) into perspective. In Greek, there are two types of *na*-particles. One is the subordinating *na* which combines with the subjunctive mood and typically translates to the English infinitival *to* (e.g., Roussou 2009). The other is the deictic *nà* which has a presentative meaning and roughly corresponds to the English *here is/are*, or the Italian *ecco*, or the French *voilà*, etc. (e.g., Zanuttini 2017 for a cross-linguistic discussion, not including Greek). Unlike the subordinating *na*, the deictic *nà* is phonetically stressed; hence, the accent on ‘à’ in (1), which I use throughout as a visual aid to distinguish between the two types of *na* (adopted from Tsoulas 2017). Now, what is particularly relevant to our present discussion is that typically *nà* combines only with DPs that yield ‘individual’ readings (e.g., Tsoulas 2017; I take it at face value that ‘individual’ and ‘referential’, mentioned previously, are synonyms). This is shown in (2), where *nà* takes the proper name as its complement.

- (2) *Nà o Janis.*
 here-is the.NOM John.NOM
 “Here is John.”

So, on the typical observation that *nà* may combine only with DPs, the grammaticality of (1) comes as a surprise, because *wh*-clauses are standardly taken to be CPs and not DPs (Chomsky 1977, onwards). One might assume that the *wh*-clause in (1), unlike the DP in (2), is not a *prima facie* complement to *nà*, but is juxtaposed. However, I will show that the *wh*-clause in (1) is indeed complement to *nà*, just like the DP in (2). In fact, I will show that *nà* may only combine with sentential complements that are typically taken to be DPs and not CPs. On this observation, the conclusion that the *wh*-clause in (1) must map to some kind of nominal constituent, despite its clausal status, is inescapable. The nominal status of the *wh*-clause is also evident by the English translation in (1), which takes the *wh*-phrase to map to a DP. In this paper then I wish to show what the syntax of the *wh*-clause is so that it can serve as the nominal complement to the presentative *nà*, and how this syntax maps to the corresponding semantics.

Before we proceed with the relevant empirical facts, a couple of clarifications is in order: The main goal of this paper is to discuss what the combination with *nà* suggests for the external syntax of a *wh*-clause, in cases like (1), not what the possibility of combining with a *wh*-clause suggests about the properties of *nà*, *per se*. However, I hasten to note that, in order to explore the distribution of *wh*-clauses with *nà*, I will examine the distribution of *nà* with clausal elements more generally, inevitably taking the discussion about *nà* a step further, and thus contributing to the literature on the presentative *nà*. Moreover, an important question in the context of this literature is if the two types of *na* (i.e., subordinating and deictic) represent two distinct lexical items (e.g., Joseph 1981; Tsoulas 2017) or reduce to one and the same lexical item that can bear two (distinct) functions (e.g., Roussou & Tsangalidis 2010). I will have nothing insightful to contribute to this debate, but some things I will say about *nà* may bear on this issue.

The rest of the discussion unfolds as follows: in section 2, I present the relevant facts, while section 3 elaborates on the nominal status of the *wh*-clause. In section 4, I maintain a recent approach to the syntax and semantics of *nà* and extend it to cases where *nà* takes a clausal complement. Section 5 is the conclusion.

2. The ontology of *nà*-complements

As already mentioned in the introduction, typically, *nà* may combine with any type of nominal element. To be sure, apart from the proper name in (2), we have the range of possibilities in (3), where *nà* may take as complement a definite DP (cf., (3a)), an indefinite DP (cf., (3b)), a bare plural (cf., (3c)), or a mass noun (cf., (3d)).

- (3) a. DEFINITE:
 Nà to aftokinito.
 here-is the.NOM car.NOM
 “Here is the car.”
- b. INDEFINITE:
 Nà ena oreo forema.
 here-is a.NOM nice.NOM dress.NOM
 “Here is a nice dress.”
- c. BARE PLURAL:
 Nà falenes.
 here-are whales.NOM
 “Here are whales.”
- d. MASS:
 Nà zaxari.
 here-is sugar.NOM
 “Here is sugar.”

Now, as also mentioned in the previous section, a standard assumption about *nà* is that it can only take nominal complements. As I will show shortly, this is not true: *nà* may also combine with clausal constituents. While our focus is on the association of *nà* with *wh*-clauses, let us place this association in the broader distribution of *nà* with the major clause-types.² To this end, we will consider, in turn, three families of clause-types: (i) independent (or main) clauses; (ii) dependent (or embedded) clauses; and (iii) relative clauses. First, witness the independent clauses in (4).

- (4) a. DECLARATIVE:
 Nà *(#) fevji o Nikos.
 here-is leaving-3SG the.NOM Nick.NOM
literally: “Here is that Nick is leaving.”
- b. YES/NO-QUESTION:
 *Nà (#) fevji o Nikos?
 here-is is-leaving.3SG the.NOM Nick.NOM
literally: “*Here is Nick leaving?”
- c. WH-QUESTION:
 *Nà (#) pjos tha erthi me mas?
 here-is who.NOM will come.3SG with us
 “*Here is who will come with us?”

(4a) shows that *nà* is grammatical with a declarative clause provided that there is a short pause (indicated here with the ‘#’ sign) between *nà* and the clausal constituent. In passing, note that a short pause is also possible between *nà* and its DP-complements in (3), but crucially for our present purposes, the pause in (3) is optional, while in (4) is obligatory. Without entering into any technical details here, it suffices to point out that the optionality of the pause in (3) and its obligatoriness in (4) suggest that the association of *nà* with the DPs in (3) is open to either subordination or adjunction, while the association of *nà* with the declarative clause in (4a) is open only to adjunction, excluding subordination. Moreover, the fact that the short pause rescues the association of *nà* with the

² A very interesting case of clausal association with *nà* is with *pu*-factives, as in *Na pu efighe noritera telika* (“Here that s/he left earlier eventually”). These cases are discussed in Χριστίδης (1985). A reviewer points out that such constructions have the non-pause pattern of the ‘*nà*+DP-complement’ constructions discussed in the present paper. I totally agree with the reviewer. As the reviewer also comments, an approach to these constructions phrased in terms of the analysis presented here would bring them closer to the analysis of the ‘*nà*+Pseudo-relative’ construction, examined in the present paper under (7). Tempting as this extension of the present analysis might be, I would have to leave it aside for now, due to space limitations.

declarative clause means that the former is not immune to the latter clause-type. However, the same cannot be said about the interrogative clause-type: as shown by the downright ungrammaticality of (4b) and (4c), irrespective of the presence of a pause, *nà* cannot combine with an interrogative clause, be it a yes/no-question (cf., (4b)) or a wh-question (cf., (4c)). This is an important observation because (4c) forms a minimal pair with (1): the grammaticality of (1), in the light of the ungrammatical (4c), says clearly that the wh-clause in (1) is actually subordinate to *nà* and has no interrogative interpretation, despite the presence of the wh-phrase (a point I return to below). Otherwise, the wh-clause should obligatorily yield a question reading, leading to an ungrammatical result, as in (4c). To clarify this point further, the ungrammaticality in (4b,c) says that *nà* is not compatible with certain clause-types (whatever the form of association is between *nà* and the clause attached to it). This brings us to the distribution of *nà* with dependent clauses, which is the issue we turn to next.

Consider the distribution in (5), where (1) is repeated as (5c) for convenience.

(5) a. DECLARATIVE:

*Nà oti fevji o Nikos.
 here-is that leaving-3SG the.NOM Nick.NOM
literally: “Here is that Nick is leaving.”

b. YES/NO-QUESTION:

*Nà an fevji o Nikos.
 here-is if is-leaving-3SG the.NOM Nick.NOM
literally: “Here is if Nick is leaving.”

c. WH-CLAUSE:

Nà pjos tha erthi me mas.
 here-is who.NOM will come-3SG with us
 “Here is the person that will come with us.”

(5a) and (5b) say that *nà* is ungrammatical with both a declarative *that*-clause and an interrogative *if*-clause.³ On the other hand, as already shown, *nà* may combine with a wh-clause only if the latter has a non-interrogative reading (cf., (5c)).

The observation just made that *nà* is incompatible with a declarative *that*-clause, but may combine with a wh-clause that yields a non-interrogative interpretation, becomes even more transparent in (6) ((6a) is repeated from Roussou 2022, 202: (8)).

(6) a. Ematha pos jirizi o troxos.
 learned.1SG how/that spins.3SG the.NOM wheel.NOM
 (i) “I learned how the wheel spins.”
 (ii) “I learned that the wheel spins.”

b. Nà pos jirizi o troxos.
 here-is how/*that spins.3SG the.NOM wheel.NOM
 (i) “Here is how the wheel spins.”
 (ii) “*Here is that the wheel spins.”

As is indicated in the glossing in (6a), and the corresponding English translations in (i) and (ii), the wh-phrase *pos* is ambiguous between a wh-reading and a *that*-reading, the latter being similar to *oti* in (5a) above. Contrary to (6a), in (6b), where the wh-clause is combined with *nà*, there is no such ambiguity, the wh-reading being the only available option.

Turning to the third, and last, family of clause-types, consider the distribution of *nà* with relative clauses, as in (7).

³ For a similar observation regarding the incompatibility of *nà* with *that*-clauses in Greek, see Tsoulas (2017: 40).

- (7) a. RESTRICTIVE RELATIVE CLAUSE (RRC):
 Nà o jatros pu me esksetase.
 here the.NOM doctor.NOM that me.ACC examined.3SG
 “Here is the doctor that examined me.”
- b. NON-RESTRICTIVE RELATIVE CLAUSE (RC):
 Nà i Maria pu psaxnis.
 here the.NOM Mary.NOM that looking-for.2SG
 “*Here is Mary that you’re looking for.”
- c. FREE RELATIVE (FR):
 *Nà ópjos erxete.
 Here-is whoever.nom is-coming.3SG
 “*Here is whoever coming.”
- d. PSEUDO-RELATIVE (PR):
 Nà i Maria / tin pu erxete.
 here the.NOM Mary.NOM her that is-coming.3SG
literally: “Here is Mary / she is coming.”

Let us describe those aspects of the distribution above that are relevant to the present discussion, without entering into much detail, due to space limitations. With the exemption of FRs (cf., the ungrammaticality of (7c), to which I return shortly), *nà* is compatible with RCs (cf., (7a) & (7b)), including PRs (cf., (7d)). Note that (7a) and (7b) become ungrammatical if we substitute *pu* (‘that’) with a *wh*-phrase.⁴ This might be an issue of register (as Dimitris Michelioudakis, p.c., has pointed out to me). Whatever the reason of this ungrammaticality may turn out to be, it is clear that it is not related to the DP-status of the clause (i.e., both *pu*- and *o opios*- relatives are DPs). For brevity, notice that (7d) bears the typical signature of PRs (see, e.g., Angelopoulos 2015): (i) the head may cliticize; (ii) the part with *pu* yields an eventive interpretation; and (iii) *pu* cannot be substituted by a *wh*-phrase (irrespective of *nà*). Now, returning to (7c), at first sight, the ungrammaticality of *nà* with FRs is quite unexpected if the present claim that *nà* takes only DP-complements is true. Yet, as a reviewer correctly points out (albeit to raise a different issue related to the analysis proposed here), this ungrammaticality shows further that the DP that *nà* may combine with must bear a referential interpretation. This is supported by the fact that, outside the domain of relative clauses, *nà* is quite bad with non-referential DPs, such as **na tipota/kanenas* (‘*here is nothing/no one’) (the examples are the reviewer’s).

Taking stock, let us reflect on the pattern that the distribution of *nà* with clauses suggests, putting the syntax of *wh*-clauses into perspective, with the aim to return to it in more detail in the next section. From the discussion revolving around independent (cf., (4)) and dependent (cf., (5)) clauses, it becomes clear that a *wh*-clause following *nà* surfaces in a position reserved for complements. This is because no short pause is permitted between *nà* and the *wh*-clause. This point becomes even more obvious by the ungrammaticality of (8), which is the *wh*-in-situ counterpart of (5c).

- (8) *Nà tha erthi me mas pjos.
 here-is will come.3SG with us who.NOM
intended: “Here is the person that will come with us.”

As Vlachos (2012) shows, Greek has an optional *wh*-in situ strategy, alongside *wh*-fronting. *Wh*-in-situ is generally available, apart from cases where the *wh*-clause is selected by a predicate and the embedded C that introduces the *wh*-in situ clause is not independently realized by a distinct lexical item (see also Vlachos & Chiou 2020 and Vlachos 2024 for more recent discussions of the *wh*-in-situ strategy). Within this frame, the ungrammaticality of (8) means that the relation between *nà* and a *wh*-clause boils down to a typical predicate-argument configuration.

⁴ At least, to my native ears. I mention this because a reviewer takes the examples with the full *wh*-phrase to be grammatical in this context, albeit a bit degraded in their acceptability, and at any rate, not more degraded than in other contexts where relative clauses are introduced with full *wh*-phrases.

Next, it is a standard observation in the literature on relative clauses that they have the distribution of DPs and not CPs. On this observation, it is quite expected that relative clauses may combine with *nà*, on a par with the DPs in (3). What is unexpected is the association of a wh-clause with *nà* (as in (5c)), on the typical assumption that a wh-clause is a CP and not a DP. So, the question that the following section addresses is why *nà* treats a wh-clause as a nominal element.

3. Labeling the wh-clause

The empirical evidence considered in section 2 suggests that *nà* imposes a certain restriction on its complement, namely, that it is nominal (and referential, but I will put this aside in this section and return to it in section 4). Restricting attention to wh-clauses, this requirement means that the external syntax of a wh-clause bears nominal properties, otherwise a wh-clause could not serve as complement to *nà*, contrary to facts. In fact, this is a run-of-the-mill requirement, as (9) shows.

- (9) a. *Nà pjos tha erthi me mas.*
 here-is who.NOM will come.3SG with us
 “Here is the person that will come with us.”
- b. *Nà pu tha pame djakopes.*
 here-is where will go.3PL on-vacation
 “Here is where we will go on vacation.”
- c. *Nà pote tha erthi o Nikos.*
 here-is when will come.3SG the.NOM Nick.NOM
 “Here is when Nick will come.”
- d. *Nà pos jirizi o troxos.*
 here-is how spins.3SG the.NOM wheel.NOM
 “Here is how the wheel spins.”
- e. *Nà jati afksanode i times.*
 here-is why increase.3PL the.NOM prices.NOM
 “Here is why the prices go up.”
- f. *Nà poso psilo ine to ktirio.*
 here-is how tall is.3SG the.NOM building.NOM
 “Here is how tall the building is.”

As we may observe, a wh-clause can serve as complement to *nà* irrespective of the grammatical category that the wh-phrase heading the clause may belong to. To be sure, the wh-phrase may be either nominal (cf., (9a)), or adverbial (cf., (9b-e)), or adjectival (cf., (9f)) (for convenience, (9a) and (9d) are repeated from (5c) and (6b) respectively).

One issue, then, is to show how the external syntax of a wh-clause maps to a nominal constituent, despite its clausal status. Another issue is to show that the wh-phrase introducing the clause has a nominal core, irrespective of the grammatical category that the wh-phrase may belong to (Noun, Adverb, or Adjective). And yet another issue to be considered concerns the internal syntax of a wh-clause, and more in particular, the association of C with the wh-phrase, which must not yield a ‘question’ reading, otherwise *nà* could not take the wh-clause as its complement, contrary to facts (as we may recall from section 2, *nà* may only combine with declarative clauses). I elaborate on these issues below, in two steps. Specifically, first, I maintain a recent approach to the morphosyntax of Greek wh-phrases, which assumes that a wh-phrase, irrespective of its grammatical category, has a nominal core. Next, I capitalize on this assumption by arguing that the nominal core of a wh-phrase, in association with the properties of C, assigns a D-label to the clausal constituent, thereby creating the illusion of a nominal element, despite the clausal status (anticipating the discussion that follows, I take ‘labeling’ in the technical sense of Chomsky 2013).

To begin with the morphosyntax of Greek wh-phrases, recently, Roussou & Vlachos (2021) revisit the internal structure of wh-phrases, in the light of their morphological assimilation to existential indefinite pronouns (e.g., *kajjos* ‘someone’) and free relative pronouns (e.g., *opjos* ‘whoever’). The

authors propose that wh-phrases are not intrinsically interrogative, but bare indefinites that provide the base for the formation of the other two quantificational elements (see also Roussou 2022). More precisely, the authors propose that the morphosyntax of a wh-phrase assumes, at least, two main components, as illustrated in (10), with *pjos* ‘who’.

(10) [D pj [φ os]]

As we may observe, the wh-phrase is composed of an indefinite D-stem, realized by the morpheme *p(j)*, and an inflectional affix, where φ are the phi-features (gender, person, number) attached to the stem, also marked for animacy, and ranging over ‘person’. So, in (10), *pjos* follows the nominal declension for adjectives, carrying the features masculine, singular, nominative. In case there is an NP-restrictor, as in, e.g., *pjos fititis* ‘which student’, modification is at the phrasal level, and the relevant morphosyntax is [DP [D pj [φ os]] [NP fititis]].⁵ In the case of wh-adverbials, specifically *pu* (‘where’), *pos* (‘how’), *pote* (‘when’), there are no phi-features, and hence, no NP-restrictor. The *p*-stem is associated with affixes that map to the descriptive content of ‘place’ (-*u*), ‘manner’ (-*os*) or ‘time’ (-*ote*) and can be taken as manifestations of oblique cases (see also Angelopoulos 2019 on the morphosyntactic decomposition of *pu*). In short, the morphosyntax in (10) says that a wh-phrase, irrespective of its nominal, adverbial or adjectival reading, is headed by an indefinite D-stem. This assumption feeds the discussion that follows, which concerns the label of the wh-clause.

To make things more concrete, let us describe the labeling problem we are dealing with, in the context of (11), concentrating on the parts of the structure that concern us here ((11) represents (9a)).

(11) *Nà* [pjos [C tha erthi me mas]].

The problem with (11) is that while *nà* may select a wh-clause, neither does it select CPs nor questions (as we saw in section 2). This means that the wh-clause in (11) is neither a CP nor a wh-question, otherwise (11) would be ungrammatical, contrary to facts. So, the question I wish to address below is what the wh-clause in cases like (11) amounts to.

Let us put the derivation of the wh-clause in (11) into perspective, by first considering the derivation of an embedded wh-question, like (12), under standard minimalist assumptions.

(12) Anarotjeme [CP_[Q] pjos [C_[Q,EPP] tha erthi me mas <pjos_[Q]>]].
wonder-1SG who-NOM will come-3SG with us
‘I wonder who will come with us.’

Focusing on the parts of the derivation in (12) that are most relevant to our present concerns, and glossing over several technical details, let us follow Chomsky (2013) in assuming that the wh-phrase *pjos* externally merges (EM) in the internal argument position of the predicate, carrying a Q-feature (Chomsky 2013 maintains that Q is the most prominent feature of wh-phrases, in line with Cable 2010, among others). As the derivation proceeds (in a *phase*-based fashion, but nothing hinges on this at present; see Chomsky 2008), C EMs with a Q-feature and a generalized EPP-feature (or Edge-feature), and probes for an appropriate Goal. C and the wh-phrase Agree, and the latter internally merges (IM) to [Spec,CP] due to the former’s EPP. After IM, the wh-phrase forms a copy with the identical inscription in the argument position, under *FormCopy* (see Chomsky et al. 2023; copies are enclosed here in angle brackets for ease of illustration). At LF, the (structurally) higher copy translates to a quantifier that binds the variable discharged by the (structurally) lower copy; hence, the ‘wh-question’ interpretation (on the ‘Duality of Semantics’, see Chomsky et al. 2023 for a more recent discussion). At PF, only the higher copy is spelled-out. In line with Chomsky (2013), the label of the wh-clause is the most prominent feature shared between C and the wh-phrase; that is, Q. Successful labeling ‘freezes’ the wh-phrase at [Spec,CP] (see also the relevant discussion in Rizzi 2006, 2010

⁵ See also Déchaine & Wiltschko (2002) for a similar account of the composition of pronouns.

and Chomsky 2015), and forms the right complement for the rogative predicate *anarotjeme* (‘wonder’), which typically selects questions (see e.g., Grimshaw 1979; Lahiri 2002).

In Chomsky’s (2013) approach, a wh-clause bears a Q-label. However, this is not the right label for the wh-clause in (11), because the interpretation of the wh-clause is clearly not interrogative, and, as we may recall from section 2, the complement of *nà* must be a ‘nominal’ element (that is, either a noun or a clause with nominal properties). In other words, the grammaticality of (11) suggests that the wh-clause carries a D-label. The way to explain this, I argue, lies in a proposal made by Roussou & Vlachos (2023) that, in embedded wh-clauses (at least), C is endowed with a generalized Op(erator) feature. I want to elaborate on this idea further, bringing it together with the morphosyntactic analysis of wh-phrases as D-elements, discussed above. This is illustrated in (13) (recasting (11)).

(13) *Nà* [_{CP}_[D] *pjos* [_C_{[Op_[D],EPP]} *tha erthi me mas* <[_{DP} *pjos*]>].

I take ‘generalized’ Op to mean that Op may take any quantificational force. That is, Op is unvalued, and in order to get a value, it must be associated with an element that can serve as a quantifier. So, Op itself is not a quantifier, but can assign quantificational properties to the element that it may be associated with. Within the range of possible values that Op can take is D, in that DPs can serve as quantifiers (as is typically the case with various types of DPs). So, Op probes for and agrees with the D-part of the wh-phrase, namely, *p*. As we may recall, under the morphosyntactic analysis of wh-phrases discussed above, *p* is not the realization of a (wh-)quantificational feature, but the manifestation of an indefinite D-stem. What turns the indefinite wh-phrase to a quantifier is its association with C, under Agree. Furthermore, EPP on C requires IM of *p* with Op. But since *p* is a bound morpheme, it must pied-pipe the entire wh-phrase to [Spec,CP], giving rise to the effect of ‘displacement’. Just like in (12), in (13), *FormCopy* applies and successful labeling results in the ‘freezing’ of the wh-phrase to [Spec,CP]. Notice, however, two crucial differences between the derivation in (13) and the one in (12): (i) the most prominent feature shared between C and the wh-phrase is D, so the wh-clause bears a D-label (not a Q-label, as in (12));⁶ and (ii) the operator-variable ‘wh-chain’ created in (13) under *FormCopy* is not interpreted as a ‘wh-question’ (unlike in (12)). In the following section, I consider the interpretation assigned to the wh-chain in (13), in more detail. Here, it suffices to point out that the wh-clause maps to a nominal element with a non-interrogative reading, which is the right complement to *nà*.

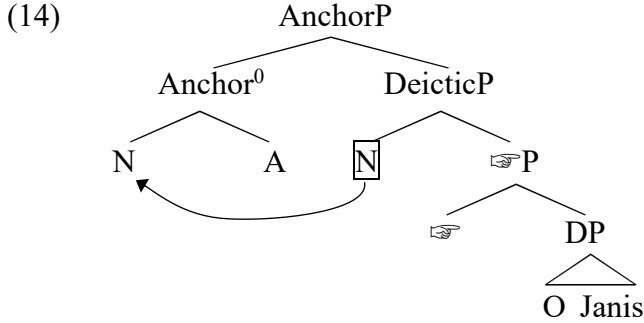
Before I end this section, I would like to briefly speculate on an analytical direction the derivation in (13) seems to point at. That is, while the derivation in (12) does not seem to extend to (13), the one pursued in (13) may extend to (12), provided that: (i) the ‘question’ reading of the wh-clause derives from the lexical semantics of the matrix predicate, and not from the internal syntax of the wh-clause; and (ii) predicates like *anarotjeme* (‘wonder’) may combine with nominal elements of some sort. So, if the analysis of (13) could be extended to (12), under the current line of theorizing, the syntax of the wh-clause in (12) would be identical to the one in (13), with both CPs been labelled by the D-property of the wh-phrase. The only difference between the two clauses would lie in the interrogative reading assigned to the wh-clause in (12) by the lexical semantics of the matrix predicate *anarotjeme* (wonder). If such an approach to (12) could be sustained, it would raise quite interesting implications vis-à-vis the formation of (embedded) wh-clauses, clause-typing, and complementation, which far exceed the scope of the present paper, and I will have to leave to future research.

⁶ An analogue of this idea can be found in the T-domain, where the ‘edge’ feature of T can be understood as the presence of a D-property (Chomsky 1995). So, Agree between T and the external argument of *v* (which is a DP) results in a shared D-property, which is interpreted on T (many thanks to Dimitris Michelioudakis, p.c., for reminding me of this analogy). Along similar lines, it is worth pointing out that the D-property in (13), in the text, is not exactly shared between C and the wh-phrase, because C itself bears no D-property before agreement with the wh-phrase. So, the D-property of C is supplied by the wh-phrase, and this D-property defines the label, on the current assumption that C’s Op is a feature in need of a value. This suggests that the ‘moved’ element, and not the target of movement, may provide the label. I will not expand on this very interesting issue, but it is not something uncommon in natural language syntax, as Cecchetto & Donatti (2015) show.

4. The *wh*-clause as complement to *nà*

In this section, I show that the way *nà* combines with a *wh*-clause, like in (13), or any clause with nominal properties, for that matter, does not differ from the way *nà* combines with an ordinary DP. To this end, I extend Tsoulas' (2017) recent treatment of *nà* with DP-complements to cases of *nà* with clausal complements.

To begin with, consider the syntax in (14) (modelled on Tsoulas' 2017, (38)).



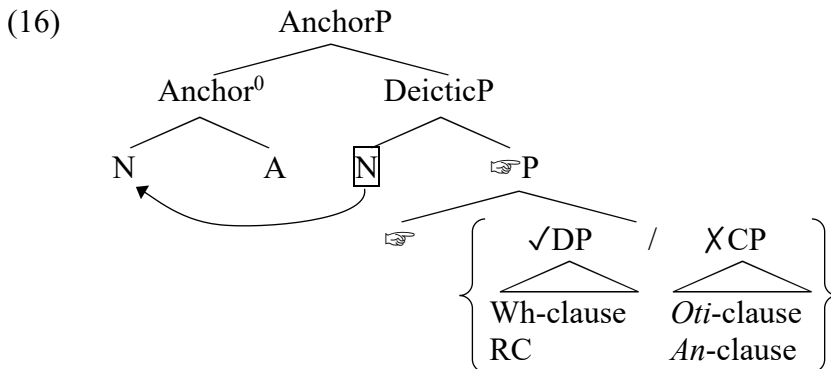
(14) says that *nà* is decomposed into two heads, an anchor *A*, and a deictic morpheme *N*, which is a relation between a pointing gesture ☞ and the relevant anchor *A*. *N* moves to *A* forming *NA* [*nà*], while the pointing gesture takes the DP *o Janis* ('John') as its argument.

The semantics of *nà* is given in (15).

- (15) a. $\llbracket \text{☞} \rrbracket = \lambda x. \lambda e. \text{☞}(x), e$
 b. $\llbracket N \rrbracket = \lambda w \lambda y \lambda Z. \text{Anchor}(\text{☞}, y) \text{ in } w$
 c. $\lambda w. \text{Anchor}(\text{☞}, x) \text{ in } w$

The gesture head (☞) is a one-place function, maps to a demonstrative (event), and takes an individual *e* as its argument (cf., (15a)). The deictic *N* introduces the subject or the anchor of the pointing (much like *v* introduces the external argument). In short, the *N* morpheme is a relation between an individual pointed at and the anchor of that pointing (cf., (15b)). The *A* morpheme introduces the relevant anchor as an individual, and when the anchor is introduced, the result is (15c) ((15b), (15c) and (15d) are repeated from Tsoulas 2017, (45), (46) and (47) respectively).

Assuming the syntax and semantics of *nà* in (14) and (15), respectively, we expect that only clauses which function as 'DPs' and range over 'individuals' can serve as complements to *nà*. In line with the empirical evidence presented in section 2, this expectation is borne out with *wh*-clauses and RCs, but not with *that/if*-clauses (and FRs, to the extent that referentiality is at stake), as illustrated in (16).



An implication (16) raises, whose more detailed discussion I will put aside at present, otherwise it would lead us too far afield, is the following: Recent approaches to clausal complementation in Greek argue that the particles *oti* ('that') and *an* ('if') are not *prima facie* 'complementizers', but pronominal elements that project a nominal layer on top of the clausal constituent (see, e.g., Roussou 2010, 2022). If so, the question that (16) raises is why *oti*-clauses and *an*-clauses cannot combine with *nà*, just as

wh-clauses and RCs do. A potential answer might be that *oti* and *an* range over propositional variables (see Roussou 2010 for further discussion), while the semantics of *nà* require a complement that discharges an individual variable.

4. Conclusion

To conclude, in this paper, I offered novel empirical evidence from Greek which shows that the deictic *nà*, in its presentative function, may take a wh-clause as its complement. On the standard observation that *nà* combines with DPs, and not CPs, its distribution with wh-clauses calls for a re-evaluation of the ‘external’ syntax of wh-clauses. The analysis put forward capitalized on two recent treatments of the morphosyntax of wh-phrases and the feature makeup of C. More in particular, the idea is that a wh-phrase is an indefinite D-element that turns to a quantifier under association with C’s unvalued Op(erator) feature (whose range of values includes D). This association results in a D-label (for the wh-clause) that is the appropriate complement to *nà*. Given that a wh-clause reduces to a DP, we expect that it patterns with other DP-clausal complements of *nà*, including relative clauses, and that it differs from *that/if*-clauses. The analysis pursued here has far-reaching implications about the syntax and semantics of wh-clauses, in particular, and clausal complement selection, more generally.

References

- Adger, David, & Josep Quer. 2001. The syntax and semantics of unselected embedded questions. *Language* 77, 107–133.
- Angelopoulos, Nikolaos. 2015. Explorations of (Greek) Pseudo-relatives. UCLA, MA dissertation.
- Angelopoulos, Nikolaos. 2019. Complementizers and prepositions as probes: Insights from Greek. UCLA, doctoral dissertation.
- Cable, Seth. 2010. *The Grammar of Q: Q-Particles, Wh-movement and Pied-piping*. Oxford: Oxford University press.
- Cecchetto, Carlo and Caterina Donatti. 2015. (Re)labeling. MIT Press.
- Chomsky, Noam. 1977. On wh-movement. In *Formal Syntax*, ed. by Peter W. Culicover, Thomas Wasow and Adrian Akmajian, 71–132. New York: Academic Press.
- Chomsky, Noam. 1995. *The Minimalist Program*. MIT press.
- Chomsky, Noam. 2008. On phases. In *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud*, ed. by Robert Freidin, Carlos P. Otero, and Maria Luisa Zubizarreta, 133–166. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2013. Problems of projection. *Lingua* 130, 33–49.
- Chomsky, Noam. 2015. Problems of projection: Extensions. In *Structures, strategies and beyond: Studies in honour of Adriana Belletti*, ed. by Elisa DiDomenico, Cornelia Hamann, and Simona Matteini, 3–16. Amsterdam & Philadelphia, PA: John Benjamins.
- Chomsky, Noam, T. Daniel Seely, Robert C. Berwick, Sandiway Fong, M. A. C. Huybregts, Hisatsugu Kitahara, Andrew McInnerney and Yushi Sugimoto (2023). *Merge and the Strong Minimalist Thesis*. Cambridge: Cambridge University Press.
- Déchaine, Rose-Marie and Martina Wiltschko. 2002. Decomposing pronouns. *Linguistic Inquiry* 33, 409–442.
- Grimshaw, Jane. 1979. Complement selection and the lexicon. *Linguistic Inquiry* 10, 279–326.
- Joseph, Brian. 1981. On the synchrony and diachrony of Modern Greek NA. *Byzantine and Modern Greek Studies* 7, 139–154.
- Kayne, Richard S. 1982. Predicates and arguments, verbs and nouns. GLOW Newsletter 8. Tilburg.
- Kayne, Richard S. 2008. Why isn’t this a complementizer. *Functional structure from top to toe* 9, 188–231.
- Kiparsky, Paul and Carol Kiparsky. 1971. Fact. *Semantics: An interdisciplinary reader in philosophy, linguistics and psychology*, ed. by Danny D. Steinberg and Leon A. Jakobovits, 345–69. Cambridge: Cambridge University Press.
- Lahiri, Utpal. 2002. *Questions and answers in embedded contexts*. Oxford: Oxford University Press.

- Manzini, M. Rita. 2010. The structure and interpretation of (Romance) complementizers. In *The Complementizer Phase: Subjects and Operators*, ed. by E. Phoevos Panagiotidis, 167-199. Oxford University Press.
- Manzini, M. Rita and Anna Roussou. 2024. Recategorizing C. In *Rich Descriptions and Simple Explanations in Morphosyntax and Language Acquisition*, ed. by Bocci, Giuliano, Daniele Botteri, Claudia Manetti, and Vincenzo Moscati. Oxford: Oxford University Press.
- Rizzi, Luigi. 2006. On the form of chains: Criterial positions and ECP effects. In *Wh-movement: Moving on*, ed. by Lisa Lai-Shen Cheng and Norbert Corver, 97-133. Cambridge, MA: MIT Press.
- Rizzi, Luigi. 2010. On some properties of criterial freezing. In *The Complementizer Phase: Subjects and Operators*, ed. by E. Phoevos Panagiotidis, 17-32. Oxford University Press.
- Rosenbaum, Peter Steven (1967). *The Grammar of English Predicate Complement Constructions*, Cambridge, MA: MIT Press.
- Roussou, Anna. 2009. In the mood for control. *Lingua* 119, 1811–1836.
- Roussou, Anna. 2010. Selecting complementizers. *Lingua* 120, 582-603.
- Roussou, Anna. 2022. Complementizers, pronouns, and labeling. *Studies in Greek linguistics* 41, 199-209.
- Roussou, Anna and Anastasios Tsangalidis. 2010. Reconsidering the ‘Modal Particles’ in Modern Greek. *Journal of Greek Linguistics* 10, 45-73.
- Roussou, Anna and Christos Vlachos. 2021. “Pronouns: Morphosyntax and complementation”. Ms University of Patras.
- Roussou, Anna and Christos Vlachos. 2023. Wh-complement clauses and (non-)local selection. *Qulso* 9: 147-159. DOI: doi: <http://dx.doi.org/10.13128/QULSO-2421-7220-15151>
- Starke, Michal (2004). On the inexistence of specifiers and the nature of heads. In *Structures and Beyond. The Cartography of Syntactic Structures*, ed. by A. Belletti, 3, 252–268. Oxford: Oxford University Press.
- Tsoulas, George. 2017. The Common Syntax of Deixis and Affirmation. In *Order and Structure in Syntax I: Word Order and Syntactic Structure*, ed. by Laura R. Bailey and Michelle Sheehan, 281-309. Berlin: Language Science Press.
- Vlachos, Christos. 2012. Wh-constructions and the division of labour between syntax and the interfaces. University of Patras, doctoral dissertation.
- Vlachos, Christos. 2024. On the left periphery of ‘optional’ wh-in-situ: Evidence from Greek. *Qulso* 10: 79-88. DOI: <https://doi.org/10.36253/qulso-2421-7220-16567>
- Vlachos, Christos and Konstantina Balasi. 2022. Conjunctive wh-clauses as nominal complements. *Selected Papers of ISTAL24*, 818-828.
- Vlachos, Christos and Michalis Chiou. 2020. The syntax, semantics and pragmatics of ‘optional’ wh-in-situ in Greek. *Journal of Greek Linguistics* 20, 102-131.
- Zanuttini, Raffaella. 2017. Presentatives and the syntactic encoding of contextual information. In *Elements of comparative syntax: Theory and description*, ed. by Zanuttini, R., Aboh, E., Haeberli, E., Puskás, G. and Schönenberger, 221-255. Berlin: Mouton de Gruyter.
- Χριστίδης, Α.-Φ. (1985). Το δεικτικό νά. *Μελέτες για την Ελληνική Γλώσσα*, 6, 221-241.